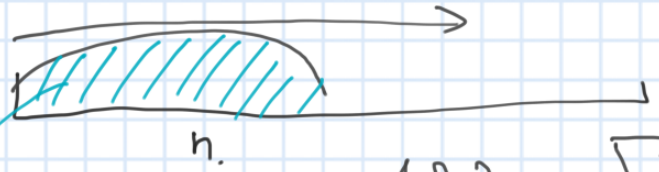
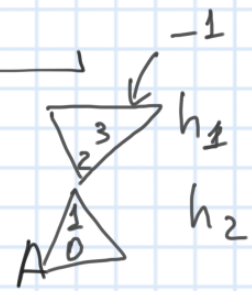


123

1 2 3 4 5 6 ←

1. 1,5 2 2,5 . . .



„В лоб"

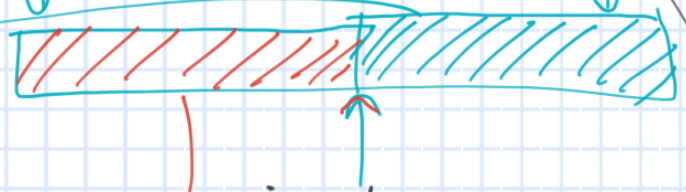
сор.т $O(n \log n)$

$n \cdot$

$O(n^2 \log n)$

1 2 3 4

```
if N1 > N2+1 :
    h2.heappush(
        h1.heappop())
```



$i = 1 \dots n$

$a[i] \leftarrow -1$.

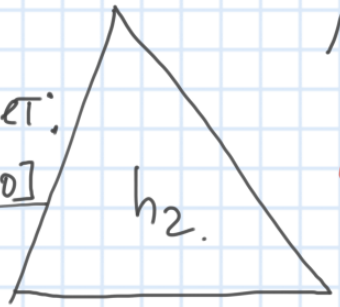
```
if N2>N1 : if a[i] ≥ h1[0] :
                h1.heappush(a[i])
        else :
                h2.heappush(a[i])
```

→ Больш. эл-ты

куча на min

N1.

$N_2+1 \geq N_1 \geq N_2$



→ мал. эл-ты.

куча на max

N2.

ОТВЕТ:

$N_1 + N_2$ - чет:

$\frac{h_1[0] + h_2[0]}{2}$

нечет

$h_1[0]$

массив $A[1..n]$, w — число.

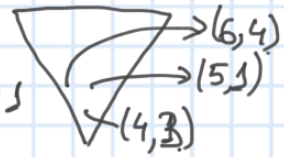
idx: 1 2 3 4

A: 5 1 4 6

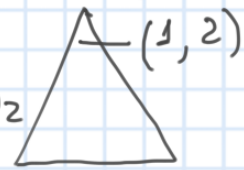
$w = 3$

A: 4 < 5

$N_1 = 2$ h_1



$N_2 = 1$ h_2



if $h_1[0][1] = i - w$:

h_1.heappop()

$N_1 -= 1$

if $a[i-w] > h_1[0][0]$:

$N_1 -= 1$.

if $a[i-w] < h_1[0][0]$:

удал. из окна. $N_2 -= 1$

2 3 4 7

$\boxed{a[i] \text{ — добав.}}$, $\boxed{a[i-w]}$

$j \le i - w \Rightarrow$ эл-т $A[j]$ не в окне $\Rightarrow$

$\Rightarrow$ удаляем, если он на верх

N_1, N_2 — кол-во „важных эл-ов" в h_1, h_2.